Chia-Yuan Wu

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Summary

Ph.D. candidate in Industrial & Systems Engineering at Lehigh University with 5 years of industry experience building production-scale software and optimization solutions. Specializing in fair machine learning, federated learning, and privacy-enhancing techniques using synthetic data and data distillation. Seeking a Summer 2026 Applied Scientist internship in Machine Learning, leveraging expertise in optimization to solve real-world challenges.

Education

Lehigh University, Bethlehem, PA
Ph.D. in Industrial and Systems Engineering — 08/2022 – Present

National Chiao Tung University, Hsinchu, Taiwan
M.S. in Industrial Engineering and Management — 09/2015 – 06/2017
B.B.A. in Management Science — 09/2011 – 06/2015

Technical Skills

- **Programming Languages & Database:** Python, C#, PL/SQL, T-SQL, Matlab, MySQL, SQL Server, Oracle
- **ML, AI & Optimization:** PyTorch, TensorFlow, Keras, scikit-learn, CPLEX, Gurobi, GLPK, AMPL
- **Libraries & Toolkits:** Pandas, Numpy, ASP.NET, VSTO, Git, CI/CD

Work Experience

Advanced Micro Devices, Inc. (AMD), Hsinchu, Taiwan — 04/2021 – 07/2022
Software System Designer (AsiaOps Planning - TW Team)

- Engineered and optimized a unified data feed solution using **MSSQL**, **C#**, and **VSTO** to automate SCM and S&OP processes, reducing user workload by **50%** and eliminating manual scheduling
- Customized and enhanced **Kinaxis RapidResponse** functionalities via **JIRA** to improve supply chain planning modules and decision-making automation, reducing user support load and streamlining planning workflows
- Delivered scalable integration of GPU/CPU planners by collaborating with cross-functional teams, managing concurrent tasks across supply chain modules, improving coordination efficiency and execution speed

AU Optronics Corp. (AUO), Hsinchu, Taiwan
Senior Software Engineer (Producion Planning Information Team) — 09/2017 – 04/2021

- Led a 3-member cross-functional team to develop **MIP-based** scheduling and resource **optimization** systems using **IBM CPLEX**, **PL/SQL**, and **C#**, reducing operational time by **90%** and replacing legacy rule-based heuristics
- Built and deployed an end-to-end optimization pipeline integrated with **SQL Server** and **Oracle DB**, boosting scheduling precision, reducing production delays, and increasing order success rate by **2X** while cutting buyer workload by **95%**
- Designed interactive **dashboards** and **KPI tracking tools** with **VSTO** and **ASP.NET**, enabling real-time decision-making and streamlined daily/weekly production monitoring

Selected Projects

Synthetic Data for Mitigating Unfairness in Collaborative Machine Learning — 06/2024 – 07/2025
LU Research Assistant — [Paper] [Code]

- Formulated a novel **bilevel optimization model** for **distributed learning**, integrating **data distillation** and **unfairness mitigation** to generate representative synthetic datasets, ensuring fairness, robustness, and privacy in server-side training
- Implemented a **single-round machine learning framework** with **synthetic data generation** and **differential privacy** to reduce client-server communication and enable scalable collaborative learning using **Python**, **PyTorch**, and **MySQL**

ATP Pegging Optimization — 09/2020 – 03/2021

- Designed and implemented an **MIP-based optimization model** using **CPLEX**, **PL/SQL**, and **ASP.NET** to allocate limited materials across product lines, converting BOM structures and business requirements into rigorous mathematical formulations
- Improved material utilization and timely order fulfillment, cutting buyer workload by **95%** and doubling the order success rate

BEOL WPS Scheduling Optimization — 09/2019 – 12/2019

- Replaced **rule-based scheduling** with a scalable **MIP-based framework** built with **CPLEX**, **PL/SQL**, and **ASP.NET** to allocate backend capacity across product lines under tight production constraints
- Reduced **WPS decommit rate** by optimizing resource utilization, enhancing efficiency and overall supply chain responsiveness

Publications [Full List]

[1] **Chia-Yuan Wu**, Frank E. Curtis, and Daniel P. Robinson. (2026). "A Bilevel Optimization Approach for Computing Synthetic Data to Mitigate Unfairness in Collaborative Machine Learning." *INFORMS Journal on Optimization.* [Paper]